Sakshi Shah

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Education

Master of Science in Mechanical Engineering *Aug 2023 – May 2025*
North Carolina State University GPA: 4.0/4.0
Bachelor of Technology in Mechanical Engineering *Aug 2017 – Jun 2021*
Savitribai Phule Pune University GPA: 8.45/10.0

Skills

Mechanical Design & CAD: SOLIDWORKS, Autodesk Inventor, AutoCAD, Engineering Drawings, GD&T, Tolerance Analysis
Prototyping & Manufacturing: SLA/FDM 3D Printing, Machining, Injection Molding, Fixturing, DFM, DFA
Engineering Analysis: ANSYS Mechanical, ANSYS Maxwell, FEA, FEMM, MATLAB/Simulink, Minitab, Python
Robotics & Automation: PLC Programming (CCW), ROS, Pneumatics, Robotic Manipulators, Electromechanical Systems
PLM & Engineering Tools: SAP PLM, Excel VBA, Power Apps, PostgreSQL

Experience

Mechanical Design Engineer **Jan 2026 – Present**
Zoetis, Durham, North Carolina

- Designed the mechanical components in a novel vaccine delivery device for a high priority R&D effort. Modeled assemblies to assess trade-offs among weight, cost, performance, and manufacturability. Considered component tolerances affecting device functionality. Components from the device were sourced, prototyped, assembled, and tested in lab conditions in preparation for further development into on-market products.
- Performed mechanical analysis on high pressure vaccine systems in order to specify size and materials suitable for the application. Used hand calculations and simulations to verify factors of safety and designed an analysis tool to perform further analysis of mechanical designs subject to the same loading conditions.
- Designed verification fixtures to position and secure the vaccine system prototype for design verification testing against functional and performance requirements. Executed tests according to established protocols, using data acquisition (DAQ) systems to measure speed, pressure, and force and characterize test repeatability and accuracy. Analyzed results to support sample size rationale refinement.
- Authored design architecture reports, and other required design control documentation to maintain traceability in accordance with the quality management system (QMS). Created design outputs and coordinated design reviews to support stage-gate deliverables.

Process Development Intern **May 2024 – Aug 2024**
Tioga Cardiovascular, Los Gatos, California

- Created new procedures and accompanying manufacturing fixturing to improve the accuracy of the catheter angle inspection in a new production line, reducing final inspection times.
- Designed SLA-printed fixtures in SOLIDWORKS and PreForm to reduce vibration during semi-automated catheter braiding. Subsequent process analysis showed a 15% improvement in yield following fixture implementation, attributed to the vibration-damping system.
- Developed a testing fixture to control the steering elements of the valve delivery system, applying Design for Manufacturing (DFM) principles to facilitate manufacturing and assembly while minimizing clinical risk.
- Designed, refined, and released engineering drawings with GD&T for mechanical components and test fixtures manufactured through machining and 3D printing. Developed controlled documents for testing and quality inspections and participated in design reviews before implementing updated product and process specifications.

Associate Engineer, Power Distribution & Controls |Python, VBA, SAP **Aug 2021 – Jun 2023**
Eaton, Pune, India

- Modeled assemblies and produced detailed component drawings for 250+ customer orders, streamlining the Engineering-to-Order (ETO) process by 34% through Continuous Improvement (CI), statistical analysis, and automation (Python, VBA, Power Apps); Six Sigma Green Belt (DMAIC) certified.
- Led Engineering Change Requests (ECRs) and Sustaining Engineering projects to resolve technical issues and address customer needs, collaborating across design, manufacturing, and procurement functions.
- Standardized Bills of Materials (BOMs) and technical documentation in SAP PLM, improving data accuracy and traceability while reducing part variations by 96%, thereby increasing mechanical assembly efficiency and improving resiliency.

- Delivered $100K in cost savings by applying Design for Assembly (DFA) and Value Analysis/Value Engineering (VAVE) to optimize sheet metal component designs by identifying solutions for performance and cost objectives.
- Created drawings, documentation, and BOMs for customers and suppliers to support fabrication and assembly of Medium voltage regulator.

Projects

Kinova Gen3 7-DOF Robot – Image-Based Visual Servoing

- Developed an image-based visual servoing controller for a Kinova Gen3 7-DOF robotic arm using MATLAB/Simulink, integrating camera-based position feedback with joint-speed control.
- Implemented image-feature tracking and resolved kinematic redundancy through a q_7 null-space strategy while maintaining fixed end-effector height and orientation.

Industrial Automation and PLC-Based Monitoring System

- Developed a PLC-based automation and monitoring system using Allen-Bradley Micro800 and Connected Components Workbench (CCW), integrating industrial devices through Modbus TCP and MQTT.
- Built a Python, Flask, and PostgreSQL-based data monitoring workflow, reducing manual intervention by 50% through automated data collection and system monitoring.

Design of an Automobile Cruise Control system

- Designed a state-feedback PID controller for a car cruise control system using control engineering principles.
- Analyzed stability using Nyquist and Root Locus plots, for an overshoot of 10% and rise time 10s.

Nonlinear Controller Design

- Designed and simulated Sliding Mode Control to achieve superior disturbance rejection with near-perfect tracking, and Adaptive Control to reduce steady-state error by 50% compared to Exact Model Knowledge.
- Evaluated computational efficiency and stability, finding that Adaptive Control balanced efficiency and performance, with 20% faster computational time.

System Integration and Control of an Autonomous Conveyor System

- Developed a Flask/PostgreSQL web application enabling remote access via Modbus TCP/IP and MQTT, for a Micro800 PLC-based color sorting system.
- Integrated sensors and actuators, reducing manual intervention by 50%.
- Integrated machine learning models (Linear, Ridge Regression, Decision Trees) for processing time prediction.

2-Pole Electro-Permanent Magnet Clamp for Workpiece Holding

- Optimized the design of an Electro-Permanent Magnetic Clamp using Magnetic Circuit Analysis (MCA) and Finite Element Method Magnetics (FEMM), achieving a vertical reluctance force of 1460 lbf.
- Validated FEMM data through 3D analysis in Ansys Electronics Desktop, ensuring accurate force calculations and identifying potential saturation effects within the EPMC design.

Equilibrium Analysis and Optimization of a Spring-Coupled Cart System

- Developed and analyzed optimization models to minimize potential energy in a spring-coupled three-cart system using Steepest Descent, Fletcher-Reeves and BFGS algorithms.
- Compared convergence rates and computational cost of gradient-based and quasi-Newton methods using MATLAB's `fminunc`, and performed sensitivity analysis on step size and gradient estimation techniques.

Simulation of Linear and Non-Linear Spacecraft Attitude Dynamics

- Re-derived and validated attitude dynamics equations from research literature using standardized notation to ensure numerical and analytical consistency.
- Simulated and compared linear vs. nonlinear models under small angle assumptions, quantifying approximation errors and defining validity bounds.

Research Engineer (Co-op Bamboochi Bicycles, Mumbai, India)

- Defined chassis design parameters and modeled the drivetrain in MATLAB to assess stability and control.
- Developed and rendered the frame design, and performed structural analysis using Finite Element Analysis (FEA) in SOLIDWORKS and ANSYS Mechanical.